

Problem Set 3

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Due: 11:59pm Central, Friday 10/03/2025. Submit a PDF to Canvas.

Instructions

This problem set will be graded on a complete/incomplete basis. Each question is worth 2 points. You will get full credit (2 points) as long as you attempt to answer the question. The questions are designed to help you prepare for the midterm on 10/08.

Problem 1

Suppose you have a random sample $\{(x_i, y_i)\}$ of size N , where each y_i is 1×1 (a scalar outcome), and each x_i is also 1×1 (a scalar regressor). If you run a linear regression of y_i on x_i without a constant, show the regression coefficient (OLS estimator) is:

$$\hat{\beta} = \frac{\frac{1}{N} \sum_i y_i x_i}{\frac{1}{N} \sum_i x_i^2}.$$

Hint: $\hat{\beta} = \arg \min_b \frac{1}{N} \sum_i (y_i - x_i b)^2$.

Solution. We want to solve

$$\hat{\beta} = \arg \min_b Q(b), \quad \text{where } Q(b) = \frac{1}{N} \sum_{i=1}^N (y_i - x_i b)^2.$$

Expanding the quadratic:

$$Q(b) = \frac{1}{N} \sum_{i=1}^N (y_i^2 - 2y_i x_i b + x_i^2 b^2).$$

Differentiate $Q(b)$ with respect to b :

$$Q'(b) = \frac{1}{N} \sum_{i=1}^N (-2y_i x_i + 2x_i^2 b).$$

Simplify:

$$Q'(b) = -\frac{2}{N} \sum_{i=1}^N y_i x_i + \frac{2}{N} b \sum_{i=1}^N x_i^2.$$

Set $Q'(b) = 0$ for minimization:

$$-\frac{2}{N} \sum_{i=1}^N y_i x_i + \frac{2}{N} b \sum_{i=1}^N x_i^2 = 0.$$

Rearrange for b :

$$b = \frac{\sum_{i=1}^N y_i x_i}{\sum_{i=1}^N x_i^2}.$$

Therefore, the OLS estimator is

$$\hat{\beta} = \frac{\frac{1}{N} \sum_i y_i x_i}{\frac{1}{N} \sum_i x_i^2}.$$

□

Remark

Since $\text{Var}(x_i) > 0$, the denominator $\sum_i x_i^2 > 0$, ensuring $\hat{\beta}$ is well-defined.

Problem 2

Prove the Frisch–Waugh theorem for the sample OLS regression coefficients by showing that the j th row of $\hat{\beta}$ is

$$\hat{\beta}_j = \left[\frac{1}{N} \sum_{i=1}^N \hat{\xi}_i^2 \right]^{-1} \left[\frac{1}{N} \sum_{i=1}^N \hat{\xi}_i y_i \right],$$

where $\hat{\xi}_i$ is the residual from an auxiliary OLS regression of x_{ji} on all the other x 's:

$$x_{ji} = x'_{(\sim j)i} \hat{\pi} + \hat{\xi}_i.$$

Hint: review lecture 4.

Solution. The OLS estimator from the full model satisfies the normal equations

$$X'(y - X\hat{\beta}) = 0 \iff \begin{bmatrix} x'_j x_j & x'_j X_{-j} \\ X'_{-j} x_j & X'_{-j} X_{-j} \end{bmatrix} \begin{bmatrix} \hat{\beta}_j \\ \hat{\beta}_{-j} \end{bmatrix} = \begin{bmatrix} x'_j y \\ X'_{-j} y \end{bmatrix}. \quad (\text{NE})$$

Let $P_{-j} = X_{-j}(X'_{-j}X_{-j})^{-1}X'_{-j}$ be the projection matrix onto the column space of X_{-j} and $M_{-j} = I - P_{-j}$ the residual-maker matrix. Premultiplying the model by M_{-j} yields

$$M_{-j}y = M_{-j}x_j \beta_j + M_{-j}u \quad \text{since } M_{-j}X_{-j} = 0.$$

Because M_{-j} is symmetric and idempotent, the OLS coefficient from this transformed regression is

$$\hat{\beta}_j = \frac{(M_{-j}x_j)'(M_{-j}y)}{(M_{-j}x_j)'(M_{-j}x_j)} = \frac{x_j'M_{-j}y}{x_j'M_{-j}x_j}. \quad (1)$$

Now consider the auxiliary regression of x_j on X_{-j} :

$$x_j = X_{-j}\hat{\pi} + \hat{\xi} \implies \hat{\xi} = M_{-j}x_j,$$

so $\hat{\xi}$ is exactly the vector of residuals from projecting x_j on *all* the other covariates. Because M_{-j} is symmetric,

$$x_j'M_{-j}y = (M_{-j}x_j)'y = \hat{\xi}'y \quad \text{and} \quad x_j'M_{-j}x_j = (M_{-j}x_j)'(M_{-j}x_j) = \hat{\xi}'\hat{\xi}.$$

Substituting these into (1) gives

$$\hat{\beta}_j = \frac{\hat{\xi}'y}{\hat{\xi}'\hat{\xi}} = \left[\frac{1}{N} \hat{\xi}'\hat{\xi} \right]^{-1} \left[\frac{1}{N} \hat{\xi}'y \right] = \left[\frac{1}{N} \sum_{i=1}^N \hat{\xi}_i^2 \right]^{-1} \left[\frac{1}{N} \sum_{i=1}^N \hat{\xi}_i y_i \right],$$

which is the desired sample Frisch–Waugh representation. □

Problem 3

We observe y_i, x_i, z_i in a sample of N observations. Assume $\bar{x} = 0, \bar{z} = 0$. Consider

$$y_i = a_0 + a_1x_i + v_i$$

estimated by OLS with coefficients $\hat{a}_0, \hat{a}_1$. Show that if $\frac{1}{N} \sum_i x_i z_i = 0$, then in the regression

$$y_i = b_0 + b_1x_i + b_2z_i + e_i,$$

the OLS satisfies $\hat{b}_0 = \hat{a}_0, \hat{b}_1 = \hat{a}_1$.

Solution. Let the residuals from the first (simple) regression be

$$\hat{r}_i \equiv y_i - \hat{a}_0 - \hat{a}_1x_i.$$

The OLS FOCs for this regression are

$$\sum_{i=1}^N \hat{r}_i = 0, \quad \sum_{i=1}^N x_i \hat{r}_i = 0. \quad (\text{FOC-A})$$

For the multiple regression, the OLS residuals are

$$e_i = y_i - b_0 - b_1x_i - b_2z_i.$$

The FOCs are

$$\sum_{i=1}^N e_i = 0, \quad \sum_{i=1}^N x_i e_i = 0, \quad \sum_{i=1}^N z_i e_i = 0. \quad (\text{FOC-B})$$

Now set $b_0 = \hat{a}_0$ and $b_1 = \hat{a}_1$. Then

$$e_i = (y_i - \hat{a}_0 - \hat{a}_1 x_i) - b_2 z_i = \hat{r}_i - b_2 z_i.$$

Plugging this into (FOC-B) gives

$$\sum_i e_i = \sum_i \hat{r}_i - b_2 \sum_i z_i = 0 - b_2 \cdot 0 = 0,$$

because $\bar{z} = 0$, and

$$\sum_i x_i e_i = \sum_i x_i \hat{r}_i - b_2 \sum_i x_i z_i = 0 - b_2 \cdot 0 = 0,$$

by (FOC-A) and the orthogonality condition $N^{-1} \sum_i x_i z_i = 0$.

The third FOC determines b_2 :

$$\sum_i z_i e_i = \sum_i z_i \hat{r}_i - b_2 \sum_i z_i^2 = 0 \implies \hat{b}_2 = \frac{\sum_i z_i \hat{r}_i}{\sum_i z_i^2}.$$

Hence with $b_0 = \hat{a}_0$ and $b_1 = \hat{a}_1$ the three FOCs for the multiple regression are satisfied (and the regressor matrix has full column rank under $\sum_i z_i^2 > 0$), so the OLS estimator for the expanded model must obey

$$\boxed{\hat{b}_0 = \hat{a}_0, \quad \hat{b}_1 = \hat{a}_1}$$

as claimed. (The coefficient on z_i is simply the OLS coefficient from regressing the first-stage residual $\hat{r}_i$ on z_i .) $\square$

Problem 4

Suppose you can observe wages, education, immigrant status, and race/ethnicity of workers.

(a) Compare the regressions:

$$y_i = \lambda_0 + \lambda_1 \text{Imm}_i + u_i, \quad y_i = \beta_0 + \beta_1 \text{Imm}_i + \beta_2 \text{Educ}_i + \epsilon_i.$$

How does $\hat{\lambda}_1$ relate to $\hat{\beta}_1$? Which assumption ensures $\hat{\lambda}_1 > \hat{\beta}_1$? (*Hint*: use omitted variable bias.)

- (b) Split immigrants into 3 groups (Asian, Hispanic, Other). Specify a regression that estimates mean log wages of each group relative to natives.
- (c) How much of the log wage gaps between immigrant groups and natives can be explained by education? Specify the regression(s) and explain how you'd answer this.

Problem 5

You have county-level data $\{y_c, x_c, z_c\}$:

- y_c : change in log wage among natives (1990–2000),
- x_c : % change in immigrant population (1990–2000),
- z_c : share of restaurants owned by Asian immigrants in 1990.

Model:

$$y_c = \beta_1 + \beta_2 x_c + u_c.$$

- (a) OLS regression $y_c = \hat{\beta}_1 + \hat{\beta}_2 x_c$. Interpret $\hat{\beta}_2$. If $\text{cov}(x_c, u_c) = 0$, show consistency.

(b) If $\mathbb{E}[u_c x_c] \neq 0$, what assumptions are needed for z_c to be a valid instrument?

(c) Suppose z_c is valid. First stage:

$$x_c = \pi_0 + \pi_1 z_c + \nu_c.$$

Reduced form:

$$y_c = \delta_0 + \delta_1 z_c + \epsilon_c.$$

Show $\delta_1/\pi_1 = \beta_2$.

(d) Do you think z_c is a good instrument? Why or why not?